

TRACT: Towards Large-Scale Crowdsensing With High-Efficiency Swarm Path Planning

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ABSTRACT

Unmanned Aerial Vehicle (UAV) sensing swarms have grown in popularity because of their advantages of low-cost, high mobility, and high maneuverability in crowdsensing. The effectiveness of UAV swarm sensing is determined by the path of each UAV in the swarm. However, it is challenging to do path planning for a UAV swarm to satisfy various data requirements in large-scale crowdsensing tasks due to the area scale and long-term planning requirements. To achieve this, we propose *TRACT*, TowARds lARge-sCaLe crowdsensing wiTh high-efficiency swarm path planning algorithm. We expand the sensing coverage problem for large-scale modeling. Furthermore, we propose a policy matrix searching technique with a simulated annealing algorithm to address the complexity of long-term planning. Our experiment shows that *TRACT* adapts to various data requirements, with the performance improved by 90% in terms of KL-divergence between the time-aggregated data value and the data requirements, while time cost for path planning was reduced by 43% compared with the previous state-of-the-art method approach.

CCS CONCEPTS

• **Computer systems organization** → **Sensor networks**; *Robotic autonomy*; • **Theory of computation** → *Random search heuristics*.

KEYWORDS

Mobile Crowdsensing; Path Planning; Simulated Annealing

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1 INTRODUCTION

With the rapid development of crowdsensing systems and unmanned vehicle platforms, Unmanned aerial vehicle(UAV) sensing swarms have become a popular field of research. UAV sensing swarms empower data collection tasks with the advantage of low-cost [1, 3], high mobility, and high maneuverability. Mounted with proper sensing systems, UAV swarms can carry out a wide variety of user-defined crowdsensing tasks. Currently, UAV sensing swarms play an essential role in various applications, including urban governance (e.g., traffic control, pollution management), wild exploration (e.g., wildfire detection, wildlife sampling) [2, 9, 10, 13, 18, 23, 31], etc.. In the mobile crowdsensing paradigm [4, 26, 28–30], UAV sensing swarms are expected to further improve the management and quality of citizen's everyday life [19].

In order to perform well in such tasks, UAV sensing swarms need to carefully plan the data collecting paths. Within a defined region, a UAV sensing swarm's sensing coverage is determined by the UAV platforms' paths and the sensors' operational range. In addition, the optimal path of UAV sensing swarms depends on data requirements. Data requirements infer the task-specific sensing coverage requirements. Take urban air pollution management as an example. On one hand, the mobile agents need to spread out to gather the air quality of different regions in the city. On the other hand, they need to focus on potential pollution sources like factories or construction sights. In many crowdsensing applications, the scale of the sensing area is too large to be simultaneously covered, and the data requirements are complicated. These constraints make long-term planning necessary for the UAV sensing swarm. In a word, to sense a large-scale area with a UAV sensing swarm, we need to plan the long-term path of each UAV agent so that the collected

data satisfy various data requirements provided by crowdsensing applications.

Various methods for swarm path planning have been proposed on a much limited scale or efficiency. Traditional robot swarm surveillance approaches used few agents to cover a large area by dividing the labor among the robots [17, 20–22]. However, with complicated data requests, static space partition would induce an unbalanced workload, thereby reducing the overall efficiency of sensing coverage. Xu and Chen *et al.* used a dynamic programming-based algorithm for path planning with respect to the data requirements [5, 27]. In this case, the complexity grows exponentially as the decision epoch increases, which makes it impossible for long-term scheduling. Liu *et al.* used deep Q learning to cover the map with a fairness metric [15]. This approach, however, suffers low efficiency on various requirements and needs to re-train the model for different data requirements. In general, existing works have poor adaption as the scale of the sensing area increases, making the UAV swarm much more limited in terms of coverage scale and sensing coverage.

This paper aims to answer the following question: **How to plan the path for a UAV sensing swarm to satisfy various data requirements in large-scale sensing tasks?** The major challenge lies in two aspects. *First, the difficulty of achieving optimal sensing coverage increases with the scale of the sensing area.* In a large-scale scenario, the UAV sensing swarm cannot simultaneously cover all the areas with data requirements. Thus, to collect the data as required, the path of the UAV sensing swarm should be balanced between the sensing coverage rate and timeliness of the measurement. *Second, the long-term decision problem is complex and hard to solve.* Given complicated data requirements, the agents need to perform long-term path planning to cover a large area with a swarm of agents. However, the combinatorial nature of the multi-agent system actions makes it intractable to enumerate all the possible long-term traces.

This paper propose *TRACT*, TowArds lARge-sCAle crowdsensing wiTh high-efficiency swarm path planning algorithm to address the challenge. To address the challenges of large area scale, we modeled the problem as an extended model originally from [27] by adding a time-aggregation data value. We expanded the modeling of the sensing coverage as the KL-divergence between time-aggregated data value and data requirements, which utilizes mobility to compensate for the operational range of the sensor swarm. This makes data collected in different decision epochs comparable to the data value within sensing coverage. To address the challenges of long-term path planning, we designed a simulated annealing-based algorithm to search for the optimal path for the UAV sensing swarm. The simulated annealing algorithm search for a policy matrix with iterative improvement strategies. This algorithm converges much faster, which enables long-term path planning with high efficiency. Our experiment shows that, within reasonable computational cost, the algorithm carries out an optimal solution.

To sum up, the main contributions of this paper are as follows:

- (1) We propose a novel formulation of the large-scale sensing coverage problem.

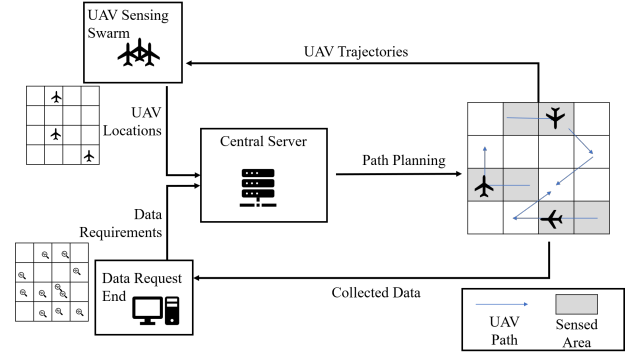


Figure 1: This figure shows the architecture of our UAV swarm sensing system.

- (2) We provide an efficient solution for long-term path planning by searching for an optimal policy matrix with simulated annealing.
- (3) We evaluate our solution based on simulation environments with physical world meanings.

The remainder of this paper is organized as follows: §2 presents the system model and problem formulations. §3 presents our algorithm design and methodology. §4 presents simulation results for performance evaluation. Finally, §5 concludes the paper.

2 PROBLEM FORMULATION

2.1 Background Definition

A typical UAV swarm sensing system includes a UAV sensing swarm, central server, and data request ends, as shown in Figure 1. The swarm of UAV sensors is responsible for sensing data in the grid map. The center server is responsible for planning the path for each UAV sensor. The data request ends are responsible for generating the data requests, which form up the data requirements. Since the area is too large for the given sensor swarm to cover, our modeling should consider the sensing coverage with data aggregation.

In general, we have K UAV agents in a quadrangular grid with the shape $M \times N$. The grid size is determined either by task-oriented criteria (*i.e.*, how fine-grained data should the UAV sensing swarm provide), or system-oriented criteria (*i.e.*, the operation range of the sensor on UAV). To collect data at decision time epochs $t = 1 \dots T$, each with dt minute. Our task is to plan the moving policy P_k for each sensor $k \in \{1, \dots, K\}$, so that the aggregated UAV agent trajectory R collects the data A that form up the data value distribution Pr , which fits the data requirements O . Each sensor would collect 1 unit of data at grid (i, j) at time t automatically. The detailed formulation is described below.

2.2 UAV Swarm Trajectory

We use $R_k \in \mathbb{R}^{M \times N \times T}$ to denote the trajectory for UAV agent k , as a three-dimensional tensor showing the appearance of a single UAV. $R_k(i, j, t)$ is the value denoting whether sensor k has shown up in grid (i, j) at time t , specifically

$$R_k(i, j, t) = \mathbf{1}(\text{location of the sensor is } (i, j) \text{ at time } t) \quad (1)$$

For simplification, we use $R_k(t)$ to denote R_k at time t . Only one grid in tensor R_k , which exists the sensor k , has the value 1, and all other grids 0. Given $R_k(0)$, a moving strategy P_k for sensor k is also a three-dimensional tensor making changes on R_k with respect to time.

$$R_k(t) = R_k(t-1) + P_k(t-1) \quad (2)$$

The possible choices of $P_k(t)$ for a given time t is limited. There are a maximum of nine possible choices, which behave like moving the sensor to its eight nearest eight-neighborhood, or staying in the same place.

2.3 Data Collection

The sensors on the UAV swarm automatically collect the sensing data. The data collection $A \in \mathbb{R}^{M \times N \times T}$ is a three-dimensional tensor, denoting how much data is collected in each grid of the map by all decision epochs. A single entry (i, j, t) of A is calculated by

$$A(i, j, t) = \sum_k R_k(i, j, t) \quad (3)$$

The data value distribution, denoted by $Pr \in \mathbb{R}^{M \times N}$ is given by aggregating the data matrix A by decision epochs and normalizing it to fit the distribution of the data requirement.

$$Pr(i, j) = f(A(i, j, t)), \sum_{i,j} Pr(i, j) = 1 \quad (4)$$

We use function $f(x)$ to denote the task-specific aggregation requirement. The input x of function $f(\cdot)$ should be a three-dimensional tensor. According to the sensing task, the value of the data could be decaying linearly or exponentially. We mainly consider the exponential decay cases, namely

$$f(A(i, j, t)) = \text{norm}(\sum_t \gamma^t A(i, j, t)) \quad (5)$$

where γ is the exponential decay factor by time, norm is the normalization term, namely $\text{norm}(\mathbf{x}) = \frac{\mathbf{x}}{|\mathbf{x}|}$. The exponential decay factor γ is determined by both the number of decision epochs and the sensing task requirements. This time-aggregation function considers mainly the most accessible data. For other tasks that care about maximal age over all cells [17], and other visual-based cases [11, 12, 24, 25], function $f(\cdot)$ needs to be redesigned.

Finally, the data requirement O is a distribution function that denotes the required data value distributions in the map. The data requirement O is generated by downstream applications. As a property of distribution functions, there are $\sum_i \sum_j O(i, j) = 1$ for all the data requirements.

We want to plan all the UAV agents' path such that the data value distribution of our collected data will need to fit the object distribution. We use KL-divergence $J(R) = D_{kl}(Pr||O)$ to measure the divergence between the collected data and the objective distribution of the data.

$$\min_{R_k, k \in 1, \dots, K} J(R) = D_{kl}(Pr||O) = Pr(i, j) \log \frac{Pr(i, j)}{O(i, j)} \quad (6)$$

2.4 Putting It Together

In general, given the initial location of each UAV, we want to find the optimal trajectory of all the UAVs, such that the value distribution of their collected data fits the target data distribution. The data distribution is given by a function related to the amount of collected data given by the data collection time. In each decision epoch, the UAV can move to its adjacent location or choose to stay in the same place. Both selections collect 1 data for the grid that the UAV stands on. A function $f(\cdot)$ indicates how the collected data is valued for task-specific requirements.

$$\begin{aligned} \min_{R_k, k \in 1, \dots, K} J(R) &= Pr(i, j) \log \frac{Pr(i, j)}{O(i, j)} \\ \text{s.t. } Pr(i, j) &= f(A(i, j, t)), \sum_{i,j} Pr(i, j) = 1 \\ A(i, j, t) &= \sum_k R_k(i, j, t) \\ R_k(i, j, t) &= 1(l_k = (i, j)) \end{aligned} \quad (7)$$

The constraints in this problem are physical mobility constraints. Considering the mobility of the UAV agents, one UAV can either move to its eight adjacent neighbourhoods or stay in the same place.

Our problem formulation shows an effort to utilize UAV swarms' mobility to compensate for their sensing coverage. In our problem, even if a sensor has not covered one grid at present, the previously collected data are still considered to be valuable. The data value function $f(\cdot)$ can be viewed as an intrinsic function of the cell age in the surveillance problem [17]. The output of $f(\cdot)$ shows how the value of the data is affected by the cell age.

3 METHODOLOGY

3.1 Overview

We proposed *TRACT* to solve the problem proposed in §2.4. Given the initial location of each UAV and data requirements, the algorithm selects optimal trajectories for each UAV agent, such that the aggregated sensing data value distribution $Pr(i, j)$ fits the data requirements $O(i, j)$ with minimized KL-divergence $D_{kl}(Pr||O)$. Since this kind of path planning problem is NP-complete[27], finding an optimized solution with a deterministic algorithm is intractable for large-scale decision epochs. Thus we consider solving the problem with random search methods like simulated annealing. To solve the problem by simulated annealing, we first model the swarm path planning problem as a policy matrix searching problem in §3.2. Then, we apply a simulated annealing algorithm to find the optimal policy matrix as a trajectory set for the UAV swarm in §3.3.

3.2 Policy Matrix

To solve the optimization problem in §2, we need to define the states for the simulated annealing algorithm. The simulated annealing algorithm searches for an optimized state as the optimization problem's solution. We mainly consider the **policy matrix**

$\mathbf{M} = [P_1, P_2, \dots, P_k]^T$ as the state of the swarm. We modeled the combinatorial policy of all the sensors in a given time scale as a state. In the policy matrix, the row number indicates the size of the UAV swarm, and the column number indicates the number of decision epochs. Each cell $M(x, y)$ in the matrix is the action that the agent x is about to carry on at decision epoch y . The optimization can therefore be viewed as a continuous search of the possible policy matrices for a sensor swarm.

We expand the policy matrix selection to all the sensors and the decision epoch. Because our simulated annealing algorithm does not seek to search the whole space but continuously improves the policy matrix from a feasible subset, the algorithm converges much faster than the previous approach. As shown in §, the simulated annealing algorithm was designed to avoid being trapped into local minimal. The search on the policy matrix is expected to converge to the global minimum.

3.3 Optimization Algorithm

By modeling the path planning into a random search problem that takes the policy matrix as the states, we can solve this problem by a simulated annealing algorithm. The complete algorithm is provided in algorithm 1

The *TRACT* algorithm first draws up a feasible solution by initializing all the mobile sensors to stay in the same place. Then, for each sampling, the system randomly chooses one agent $k \in 1, \dots, K$, then randomly chooses a decision epoch t , altering its corresponding action a_t for the given agent k . An update is applied after all the actions come directly after t , which means updating all the locations of the sensor by $t = t \dots T$, to see if some of the sensors have run out the scope. If the sensor is still in the boundary, i.e., it is still a feasible solution, then we follow the Boltzmann acceptance criterion to see if it can be updated.

The Boltzmann acceptance criterion is given in equation 8. Where B is the temperature factor. When the new policy matrix has a bigger value of J (i.e., not optimal compared with the previous policy matrix), the algorithm will accept the change within probability γ_s . With the iterations going, B will gradually reduce to a smaller value.

$$\gamma_s = e^{-\frac{J'-J}{B}} \quad (8)$$

The simulated annealing algorithm has a convergence proof for an irreducible Markov chain[14]. In our cases, each valid policy matrix forms a state in the Markov chain. If the policy matrix Markov chain is irreducible, then simulated annealing is expected to converge in the global minimum. We found that the *TRACT* converges to an optimal solution in all the cases in our experimental study. This means in practice the simulated annealing algorithm often converges to the optimal solution. Nevertheless we are working on further theory proof on the property of the policy matrix Markov chain.

4 EVALUATION

In this section, we introduce the evaluation on our algorithm's ability to achieve high-efficiency sensing coverage with scalability

Algorithm 1: Simulated Annealing Swarm Path Planning Algorithm

Input : Input: Initial location of the sensor crowd
 $R(0) = R_1(0), \dots, R_k(0)$, Initial temperature B_0 , cooling constant c

Output: Output: Improved optimal policy matrix $\mathbf{M}^*$

```

1 Initialization: set temperature  $B := B_0$ , initialize
   $J(R) = D(Pr||O)$ , policy matrix  $M = M_0$ ;
2 while true do
3   for  $j \in 1, 2, \dots, KT$  do
4     Randomly choose a row  $k \in K$ ;
5     Randomly choose a column  $t \in T$ ;
6     Randomly choose a strategy  $a' := a$ ;
7     Update policy matrix cell  $\mathbf{M}(k, t)$  by  $a'$  as  $\mathbf{M}'$ ;
8     if  $M$  is not a valid policy matrix then
9       continue;
10    end
11    Update  $R'$  by policy matrix
       $M' = [P_1, P_2, \dots, P'_k, \dots, P_K]^T$ ;
12    Calculate  $J' = J(R')$  by equation 6;
13    Calculate  $\gamma_s$  by equation 8;
14    Sample  $q \sim U(0, 1)$ ;
15    if  $J' - J \leq 0$  then
16      if  $\gamma_s \geq q$  then
17        Update  $R = R'$ ;
18      end
19    else
20      continue;
21    end
22  end
23  Update  $R = R'$ ;
24 end
25 Update temperature by  $B := cB$ ;
26 if  $|J' - J| \leq \epsilon$  then
27   break;
28 end
29 end
```

towards different swarm sizes and decision epochs. We first introduce how we set up our evaluation in section 4.1, then we present the results in section 4.2

4.1 Evaluation Setup

Our experiments are based on generated physical-based simulation data. Initial settings in our problem involve the sensing map, the decision epoch, the swarm size, the number of iterations, and the data requirements. The major setup parameters are listed below.

Experiment Setup: To evaluate our methods in different scenarios, we choose a 15×15 sized grid map and generate three different data requirements. All the agents were initially located in the bottom left of the map. The decision epoch was by default 100; the swarm size was by default 7; the number of iterations for the algorithm was by default 1500; the aggregation decay rate γ was

by default 0.99 to simulate a slow-changing field like air-pollution field. [6–8]

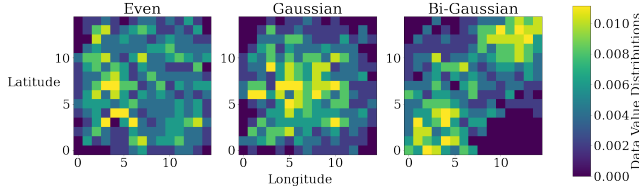


Figure 2: This figure illustrates our planned path for different data distributions.

Performance Metrics: The primary performance metric was the KL-divergence between the data requirements and the time-aggregated data value distributions. Lower KL-divergence indicates better fitting of the data requirements. To evaluate the efficiency under different requirements for *TRACT*, we have tested three distributions as the target distributions. They are (1) Even distribution: the data value should be evenly distributed. (2) Gaussian distribution: the data should focus on the center of the map. (3) Bi-Gaussian distribution: the data should focus on bottom left (the initial point of the swarm) and the top right (which forms the diagonal of the map). Besides, the running time for different algorithms to converge for different decision epochs was also evaluated. All the experiments on our server, the CPU was Intel(R) Xeon(R) Gold 6242R and the GPU was NVIDIA RTX A6000.

Baselines: We adopt two baselines to validate the effectiveness and scalability of our algorithm.

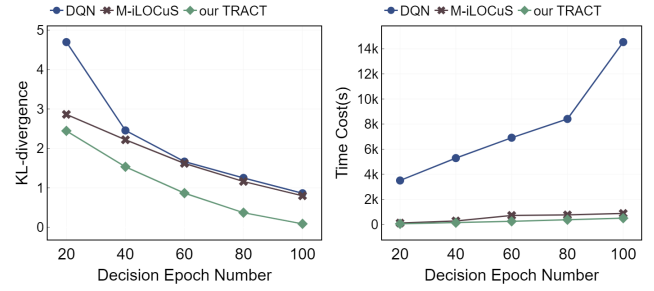
(1)**Deep Q Learning (DQN):** This method uses the deep Q learning algorithm [16] to solve the optimization problem as proposed in section 2. By comparing this method with our algorithm, we can check whether our algorithm outperforms the state-of-the-art algorithm in a large number of iterations.

(2)**Multi-step iLOCuS (M-iLOCuS):** This method is based on iLOCuS, which was originally proposed in [27]. To adapt with our problem, we use a multi-step version to overcome the curse of decision epochs. The long decision epoch was split by every 5 epochs to avoid trapping in calculating trace sets. By comparing this method with our algorithm, we can check whether our algorithm outperforms the state-of-the-art algorithm in long-term decisions.

4.2 Experiment Results

Figure 2 shows the data value distributions for complex target distributions. By optimizing the target function with a simulated annealing-based algorithm, *TRACT* successfully adapts to different requirements of data distributions. The algorithm also managed to plan a long-term path for different sensors without explicitly determining the workspace for each sensor. In the Bi-Gaussian cases, we can further see that the algorithm managed to work across the area with little profit and reach the designated area in the top right corner. This is because the Boltzmann acceptance criterion of the simulated annealing, which makes the algorithm avoid trapping into a local minimum.

Figure 3(a) shows the performance of our algorithm in different decision epochs. Our *TRACT* has outperformed all the baseline



(a) Performance Evaluation **(b) Time Cost Evaluation**

Figure 3: This figure shows the performance of our algorithm. Figure(a) shows how the KL divergence drops within different decision epochs. Figure (b) shows how the time cost responds to the scale of our problem.

algorithms, achieving the same KL-divergence with 40% fewer decision epochs. The final result of KL-divergence for *TRACT* was 0.08, much smaller than M-iLOCuS(0.79) and DQN(0.86). The reason for *TRACT* outperforming the baselines is the ability to consider the decision epoch as a whole instead of piece-wise walking, making it much easier to reach the global minimum.

Figure 3(b) shows the improvement in the time complexity brought by our method. Our method shows strong adaptivity to the number of decision epochs. The evaluation shows that the *TRACT* algorithm converges at 497s, which is nearly twice faster than M-iLOCuS(875s) and much faster than the DQN(1453s). The fast convergence of *TRACT* enables long-term path planning with affordable computational cost.

5 CONCLUSIONS

In this paper, we proposed a *TRACT* algorithm for large-scale UAV sensing swarm path planning. By formulating the problem with a time-aggregation data value, the UAV sensing swarm was able to handle large-scale sensing with data requirements. By solving the problem with policy matrix-based simulated annealing, the path planning of *TRACT* adapts to multiple target distribution requirements. Experiment results show that *TRACT* outperforms the baseline with time cost reduced by up to 43%, and managed to achieve high efficiency even for the complicated data requirements.

TRACT represents an initial effort to solve the swarm planning problems with random search and partial optimization. With further development, this algorithm will lead to a broadly applicable method for real-world large-scale sensing tasks with more strict constraints.

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